

Kai S. Yun

Legal Name: SirkHoo Yun

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RESEARCH INTEREST

My research focuses on developing **safe and scalable controllers** and deploying them on **physical robots**.
My methodology encompasses **control theory**, **deep learning**, **foundation models**, and their **intersection**.

EDUCATION

Massachusetts Institute of Technology Class of 2029 (Expected)
Ph.D. in Mechanical Engineering Cambridge, MA

Carnegie Mellon University Class of 2025
M.S. in Mechanical Engineering Pittsburgh, PA

University of California, Berkeley Class of 2023
B.S. in Mechanical Engineering, Minor in EECS Berkeley, CA

PUBLICATIONS

P: preprints, C: conference proceedings, J: journal articles, *equal contribution

- [P1] **Kai S. Yun**, Zeyang Li, Navid Azizan, “Provably Safe, Yet Scalable Reinforcement Learning”. Preprint at [arxiv.org:2606.14536](#), 2026.
- [C2] **Kai S. Yun**, Navid Azizan, “ATOM-CBF: Adaptive Safe Perception-Based Control under Out-of-Distribution Measurements”. Published in *Learning for Dynamics and Control Conference (L4DC)*, 2026. Full paper at [arxiv.org:2511.08741](#).
- [C3] Yifan Sun, Rui Chen, **Kai S. Yun**, Yikuan Fang, Sebin Jung, Feihan Li, Bowei Li, Weiye Zhao, Changliu Liu, “SPARK: Safe Protective and Assistive Robot Kit”. Published in *IFAC Symposium on Robotics*, 2025. Full paper at [arxiv.org:2502.03132](#), 2025, [Website](#).
- [C4] **Kai S. Yun**, Rui Chen, Chase Dunaway, John M. Dolan, Changliu Liu, “Safe Control of Quadruped in Varying Dynamics via Safety Index Adaptation”. Published in IEEE *International Conference on Robotics and Automation (ICRA)*, 2025. **Oral Presentation** in Atlanta, Georgia, 2025, [Website](#).
- [C5] Simin Liu*, **Kai S. Yun***, John M. Dolan, Changliu Liu, “Synthesis and Verification of Robust-Adaptive Safe Controllers”. Published in *European Control Conference (ECC)*, 2024. **Oral Presentation** in Stockholm, Sweden, 2024.
- [C6] Tianhao Wei, Hanjiang Hu*, Luca Marzari*, **Kai S. Yun***, Peizhi Niu*, Xusheng Luo, Changliu Liu, “ModelVerification.jl: a Comprehensive Toolbox for Formally Verifying Deep Neural Networks”. Published in *International Conference on Computer Aided Verification (CAV)*, 2025, [GitHub](#).
- [J7] Jack C. Harms, Ethan M. Grame, **SirkHoo Yun**, Bushra Ahmed, Leah C. O’Brien, James J. O’Brien, “Mass-independent Dunham Analysis of the $[7.7]Y^2\Sigma^+ - X^2\Pi_i$ and $[16.3]A^2\Sigma^- - X^2\Pi_i$ Transitions of Copper Monoxide, CuO”. Published in *Journal of Molecular Spectroscopy*, 2019.

HONORS & AWARDS

MathWorks Fellowship, 2026 — 2027, MIT School of Engineering
ATK-Nick G. Vlahakis Fellowship, 2024 — 2025, CMU College of Engineering
Best Master’s Research Paper Commencement Award, 2025, CMU Department of Mechanical Engineering
Conference Grants: MIT OGE (2026), CMU OGPA (2024), CMU ME (2024)

RESEARCH EXPERIENCE

Laboratory for Information & Decision Systems (MIT) Cambridge, MA
Advisor: Prof. Navid Azizan Aug 2025 — Present

- Vision-language-action (VLA) models for safe, scalable, and general-purpose robotics.
- Hard-constrained safe reinforcement learning for high-dimensional robots [P1].
- Perception-based safe control theory for out-of-distribution measurements [C2].

Robotics Institute (CMU) Pittsburgh, PA
Advisors: Prof. Changliu Liu, Prof. John M. Dolan Jun 2023 — Jul 2025

- Safe humanoid autonomy and teleoperation toolbox (*SPARK*) development (Unitree G1 platform) [C3].

- Adaptive safe controller for quadrupeds in varying dynamics (Unitree Go2 platform) [C4].
- Algorithm based on formal method for synthesizing robust-adaptive safe controllers under uncertainty [C5].
- [ModelVerification.jl](#), a Julia library containing state-of-the-art neural network verification algorithms [C6].

Berkeley AI Research Lab (UC Berkeley)

Berkeley, CA

Advisor: Prof. Koushil Sreenath

Aug 2021 — Apr 2022

- *PPO-Worst-Case*, a novel safe reinforcement learning algorithm for worst-case scenarios.
- Simulations for legged robots in safety-critical scenarios.

PROFESSIONAL EXPERIENCE

Tesla, Inc.

Palo Alto, CA

Vehicle Dynamics / Software Engineering Intern

May 2022 — Aug 2022

- Developed a correlation framework for comparing subjective vehicle evaluation and objective test data.
- Analyzed vehicle dynamics data to improve the ride, handling, and steering experiences of Models S, 3, and Semitruck.
- Tested Tesla vehicles and competitor vehicles with sensors and robots. Conducted vehicle tests on proving grounds.
- Developed automated ticketing, reporting, and logging systems utilizing internal software and corresponding API.

NeuroCore.ai

Seoul, South Korea

Reinforcement Learning Engineering Intern

Oct 2020 — Jul 2021

- Designed and developed RL training and deployment frameworks, which increased training efficiency by 84%.
- Developed simulators for Supply Chain Management (SCM) tasks for semiconductor manufacturers.
- These are currently deployed in South Korean semiconductor manufacturers, such as SK Hynix.

TEACHING & SERVICE

Robotics Institute at CMU

Pittsburgh, PA

Manager/Developer — [RoboticsKnowledgebase](#) (Funded by Google)

Oct 2024 — Jul 2025

Teaching Assistant — *F1Tenth Autonomous Racing (16.663)*

Jan 2025 — May 2025

Robot Manager, Lab Manager — [Intelligent Control Lab](#)

Jan 2025 — May 2025

Lab Server Manager — [DRIVE Lab](#)

Jun 2024 — May 2025

Research Mentor — [Robotics Institute Summer Scholars \(RISS\)](#)

May 2024 — Aug 2024

Mentee: Chase Dunaway (Undergraduate at New Mexico Tech → Ph.D. at MIT)

Mechanical Engineering Department at CMU

Pittsburgh, PA

Teaching Assistant — *Dynamic Systems and Controls (24.352)*

Aug 2024 — Dec 2024

Mechanical Engineering Department at UC Berkeley

Berkeley, CA

Teaching Assistant — *Statistics and Data Science for Engineers (ENG 178)*

Jan 2023 — May 2023

Republic of Korea Army

South Korea

Squad Leader, Sergeant, K-1 Tank Mechanic — *Combat Service Support (CSS)*

Jan 2019 — Aug 2020

Reviewer: IEEE Robotics and Automation Letters

SKILLS

Area of expertise: Humanoids, Quadrupeds, Quadrotors, ML, RL, VLA, World Models, Safe Control, Optimization

Physical Robots/Machines: Unitree G1, Unitree Go2, Pixhawk Drones, CrazyFlie, Sawyer, F1Tenth, Tanks.

Libraries: JAX, PyTorch, MJX, Gazebo, Gym, Isaac, PyBullet, Mosek, CasADi, PX4, Numpy.

Programming/OS/Tools: Python, MATLAB, Julia, C++, ROS, ROS2, Linux, Git, Docker.

Other skills: Jetson, OptiTrack, Vicon, OpenSim, SolidWorks, L^AT_EX, Simulink, Scuba Diving.

Languages: Fluent in English & Korean.

WORK AUTHORIZATION

US Permanent Resident (Green Card holder), South Korean Citizen (military service completed)